

The Moser--MacLeod Squarefree Error Estimate

Human proof and Lean correspondence

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1. Statement and notation

Let

$$Q(x) = |\{n \in \mathbb{N} : 1 \leq n \leq x, n \text{ squarefree}\}|, \quad R(x) = Q(x) - \frac{6}{\pi^2}x.$$

The formal result is

```
theorem MoserMacLeod.abs_R_lt_sqrt (x : ℝ) (hx : 0 < x) :  
  |MoserMacLeod.R x| < Real.sqrt x
```

Moser and MacLeod state this for $x \geq 1$. Lean proves it for every $x > 0$. If $0 < x < 1$, then $\lfloor x \rfloor = 0$, hence $Q(x) = 0$, and

$$|R(x)| = \frac{6}{\pi^2}x < \sqrt{x}.$$

Natural-number variables are written $\mathbb{N}$, real variables x . Thus Lean's `squarefreeCountNat N` is $Q(N)$, while `squarefreeCount x` is $Q(\lfloor x \rfloor)$.

2. Dependency diagram

- Möbius Dirichlet series and elementary Mertens bounds yield the tail estimate.
- The tail estimate and square-divisor identity yield the coarse bound for $N \geq 401$.
- The 400-cell rational certificate handles every real x with $1 \leq x < 401$.
- The finite certificate and coarse range combine to prove $|R(x)| < \sqrt{x}$ for every $x > 0$.

The certificate node consists of ordinary proof terms. It does not invoke a native evaluator.

3. Human-readable proof

3.1 The exact squarefree formula

For $n > 0$, Möbius inversion gives

$$\sum_{d^2|n} \mu(d) = \begin{cases} 1, & n \text{ squarefree,} \\ 0, & \text{otherwise.} \end{cases}$$

Summing over $1 \leq n \leq N$ and interchanging finite sums yields

$$Q(N) = \sum_{d \leq \sqrt{N}} \mu(d) \left\lfloor \frac{N}{d^2} \right\rfloor. \quad (\text{A})$$

Writing

$$\theta_{N,d} = \frac{N}{d^2} - \left\lfloor \frac{N}{d^2} \right\rfloor, \quad 0 \leq \theta_{N,d} < 1,$$

and using

$$\sum_{d=1}^{\infty} \frac{\mu(d)}{d^2} = \frac{1}{\zeta(2)} = \frac{6}{\pi^2}, \quad (\text{B})$$

we obtain the exact decomposition corresponding to equation (2) of the paper:

$$R(N) = - \sum_{d \leq \sqrt{N}} \mu(d) \theta_{N,d} - N \left(\frac{6}{\pi^2} - \sum_{d \leq \sqrt{N}} \frac{\mu(d)}{d^2} \right). \quad (\text{C})$$

Lean expands the divisor bookkeeping in (A), one of the details suppressed in the paper.

3.2 Elementary bounds

The periodic auxiliary function of Moser--MacLeod has period 30 and values in $\{0, 1\}$. Their convolution identity bounds $M(N)+1$ by a squarefree count. The formalization combines this with the convenient elementary estimate

$$Q(N) \leq \frac{3}{4}N + 1$$

and a direct proof for the small values to obtain

$$|M(N) + 1| \leq \frac{1}{8}N + \frac{3}{4} \quad (N \geq 2), \quad |M(N)| \leq \frac{1}{8}N + \frac{7}{4} \quad (N \geq 1). \quad (D)$$

For arbitrary numbers $0 \leq \theta_d \leq 1$, separating the positive and negative values of $\mu(d)$ gives

$$\left| \sum_{d \leq m} \mu(d) \theta_d \right| \leq \frac{Q(m)}{2} + \frac{|M(m)|}{2}. \quad (E)$$

This is the formal counterpart of the signed fractional-part estimate leading to equation (7) of the paper. The constants in (D) are deliberately stated as the constants actually used by this Lean proof; they are not presented as verbatim copies of the paper's displayed estimates.

3.3 The Möbius tail

Apply finite summation by parts to $\mu(d)/d^2$, using the bound for $M(d)+1$ in (D). Lean keeps both endpoint terms and proves convergence before passing to the limit. The result, combined with (E), yields for natural $N \geq 401$

$$|R(N)| < \frac{13}{16} \sqrt{N} + \frac{11}{4}. \quad (F)$$

Since $N \geq 401$ implies $\sqrt{N} > 20$, this is strong enough, after allowing for movement inside a unit cell, to prove the desired strict inequality for every real $x \geq 401$.

3.4 Why a finite certificate covers real x

For $N = \lfloor x \rfloor$ and $x \in [N, N+1)$, $Q(x)$ is constant, so

$$R(x) = Q(N) - \frac{6}{\pi^2} x$$

is affine. The proof encloses $6/\pi^2$ between $607/1000$ and $609/1000$. For each $1 \leq N \leq 400$, it verifies rational endpoint inequalities sufficient to imply

$$|Q(N) - \frac{6}{\pi^2} x| < \sqrt{x} \quad \text{throughout } [N, N+1).$$

The endpoint-to-cell passage is symbolic in `endpoint0K_sound` and `abs_R_lt_sqrt_of_lt_401`. Thus this is 400 finite integer/rational checks, not an enumeration of real inputs.

Each check is a Lean theorem made from exact reductions, primality proofs, Möbius recurrences, and `norm_num`. Rows are split into groups of 50 so that review and recompilation remain local.

4. Paper/addendum to Lean index

The paper's numbering is used where the formal development implements that step. ``Adapted constants'' means the mathematical step is the same but this repository uses the explicitly displayed bounds in Section 3.2.

- **Definition of Q, R :** Lean declaration(s): `squarefreeCountNat`, `squarefreeCount`, `R`; File: `Base.lean`; Relation: Exact
- **Identity preceding (2):** Lean declaration(s): `sum_moebius_square_divisors`, `squarefreeCountNat_floor_formula`; File: `Proof.lean`; Relation: Exact; bookkeeping expanded

- **Equation (2):** Lean declaration(s): `R_nat_eq`; File: `Proof.lean`; Relation: Exact integer specialization
- **Equation (3), rough estimate:** Lean declaration(s): ---; File: ---; Relation: Not needed
- **Equation (4), squarefree count:** Lean declaration(s): `squarefreeCountNat_le_three_quarters`; File: `Base.lean`; Relation: Same role, adapted constant
- **Auxiliary function paragraph:** Lean declaration(s): `floorMobius_eq_one`, `auxF_periodic`, `auxF_eq_zero_or_one`, `sum_mu_auxF_eq_neg_one`; File: `Base.lean`; Relation: Expanded completely
- **Equations (5), (6):** Lean declaration(s): `abs_mertensInt_add_one_le_count`, `abs_mertens_add_one_le`, `abs_mertens_le`; File: `Base.lean`, `Infrastructure.lean`; Relation: Same argument, constants (D)
- **Equation (7):** Lean declaration(s): `sum_abs_muR_eq_count`, `abs_sum_muR_mul_le`; File: `Infrastructure.lean`; Relation: Abstract weighted form (E)
- **Abel paragraph before (8):** Lean declaration(s): `finite_tail_bound`, `hasSum_mu_tail`, `abs_mu_tail_le`; File: `Proof.lean`; Relation: Endpoint and convergence details expanded
- **Equation (8):** Lean declaration(s): tail component of `abs_R_nat_lt_coarse`; File: `Proof.lean`; Relation: Adapted constants
- **Equation (9), coarse estimate:** Lean declaration(s): `abs_R_nat_lt_coarse`; File: `Proof.lean`; Relation: Adapted bound (F), cutoff 401
- **Finite verification after (9):** Lean declaration(s): `finite_endpoint_certificate`, `endpointOK_sound`, `abs_R_lt_sqrt_of_lt_401`; File: `FiniteTable.lean`, `Base.lean`, `Infrastructure.lean`; Relation: Uniform real-cell proof
- **Final conclusion:** Lean declaration(s): `abs_R_lt_sqrt_of_one_le`; File: `Proof.lean`; Relation: Exact paper domain
- **Elementary extension:** Lean declaration(s): `abs_R_lt_sqrt`; File: `Proof.lean`; Relation: Extends domain to $x > 0$

5. Lean declaration index

- `hasSum_mu_div_sq`: Mathematical job: Identity (B)
- `density_bounds`: Mathematical job: Rational enclosure of $6/\pi^2$
- `endpointOK`, `endpointOK_sound`: Mathematical job: One certified endpoint and its soundness
- `finite_endpoint_certificate`: Mathematical job: Dispatch all cells 1,...,400
- `abs_R_lt_sqrt_of_lt_401`: Mathematical job: Every real $1 \leq x < 401$
- `abs_mertens_add_one_le`, `abs_mertens_le`: Mathematical job: Bounds (D)
- `abs_sum_muR_mul_le`: Mathematical job: Weighted bound (E)
- `finite_tail_bound`: Mathematical job: Finite Abel summation with endpoints
- `hasSum_mu_tail`, `abs_mu_tail_le`: Mathematical job: Infinite Möbius tail
- `sum_moebius_square_divisors`: Mathematical job: Pointwise squarefree detector
- `squarefreeCountNat_floor_formula`: Mathematical job: Formula (A)
- `R_nat_eq`: Mathematical job: Exact decomposition (C)
- `abs_R_nat_lt_coarse`: Mathematical job: Coarse estimate (F)
- `abs_R_lt_sqrt_of_one_le`: Mathematical job: Paper's theorem for $x \geq 1$
- `abs_R_lt_sqrt`: Mathematical job: Public theorem for $x > 0$
- `sum_moebius_sq_eq_squarefreeCountNat`: Mathematical job: Bridge to the MWE's $\sum \mu(n)^2$ statement

6. Trust and reproducibility

There is no `native_decide` in the development. In particular, the finite range is not hidden behind `Lean.ofReduceBool` or the native compiler. The certificate shards are large because their proof terms expose computation to Lean's kernel.

`#print axioms MoserMacLeod.abs_R_lt_sqrt` reports only:

propext
Classical.choice
Quot.sound

No solution theorem depends on sorryAx.

For independent comparison, Challenge.lean imports only Mathlib and holds the target statement with an intentional sorry; Solution.lean repeats exactly that statement with its proof. comparator.json records the permitted axioms.